

Biologic agents for IBD come of age as host–microbe interactions emerge

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In 2024, there were major advances in inflammatory bowel disease research. Treatment strategies with biologic agents for inflammatory bowel disease matured, and gut microbiota–bacteriophage–host interaction emerged as a new frontier to target.

Among the innumerable research papers on inflammatory bowel disease (IBD) in 2024, three stood out. Two clinical studies highlighted the maturing cycle of biologic agent treatment strategies in clinical practice, and a basic science study disentangled the interaction between the gut microbiota, bacteriophages and the host immune system^{1–3}.

Prior to 2024, the most effective treatments for moderate to severe Crohn's disease could only be used after failure of 'conventional' treatment⁴. Although existing observational studies and clinical experience favoured early initiation of biologic agents there is hesitancy to this approach for most patients, partly owing to the lack of randomized controlled trials, safety concerns and costs. The results of the PROFILE study² should put this hesitancy to rest. The study clearly demonstrated the superiority of a top-down, hit hard and hit early approach with infliximab and immunomodulators in moderate to severe Crohn's disease versus a more measured approach, in which oral steroids are used to induce remission initially, only adding immunomodulators after the first flare, and infliximab in addition to immunomodulators after the second flare (accelerated step-up) (Fig. 1). More patients in the top-down group had sustained steroid-free and surgery-free survival than those in the accelerated step-up group after 48 weeks (79% versus 15%, $P < 0.0001$). Several secondary endpoints were also met. Importantly, no patients in the top-down group required surgery for IBD-related complications, whereas rates of surgery were ten times higher in the step-up group (Fig. 1).

Notably, the study² recruited patients extremely early in their disease course (median time after diagnosis of 12 days), which might explain why both arms had much higher rates of endoscopic remission than in typical IBD trials (67% in the top-down arm and 43% in the accelerated step-up group versus ~30%)⁵. Thus, early, aggressive treatment of Crohn's disease could be more effective and result in a long-term favourable disease course. It remains to be seen whether biologic treatment can be safely stopped after a longer treatment period in some of these patients.

However, IBD diagnosis cannot be established firmly immediately after onset, and some patients could have transient, self-limiting inflammation. Thus, the top-down approach could inadvertently lead to overtreatment of some patients. In PROFILE, only 15% of patients in the step-up approach were not started on immunomodulators, but in practice, overtreatment is likely to be offset by the overall favourable results.

Although small, this risk of overtreatment justifies the need for more personalized treatment, possibly based on predictive biomarkers. The use of a 17-gene transcriptional biomarker had no predictive value for response to therapy in patients enrolled in the study, but that biomarker (related to T-cell exhaustion) might not be suited for use immediately after diagnosis. Follow-up trials in appropriate populations are needed.

Also in 2024, the SEQUENCE study, a head-to-head, treat-through comparison of ustekinumab versus risankizumab showed that risankizumab was non-inferior to ustekinumab in achieving clinical remission at 24 weeks, and was superior to ustekinumab in achieving endoscopic remission at 48 weeks (Fig. 1). Patients randomly assigned to risankizumab ($n = 255$) had reduced hospitalization related to IBD or any other cause compared with those on ustekinumab ($n = 265$); all predefined secondary endpoints were reached. Importantly, serious adverse events were lower in the risankizumab group. These observations are in line with the excellent safety profile of IL-12–IL-23 and IL-23 inhibitors.

This paper highlighted three key issues. First, ustekinumab and risankizumab have a different dosing schedule: the study was open label, with the risk of bias minimized by a blinded central endoscopy reader. Importantly, high risankizumab dosages (at least four times higher than the standard dose of ustekinumab) were used. These high dosages might make measurements of drug levels or dose intensification unnecessary, and differences in dosing could partly explain the differences in outcome. Second, head-to-head studies should now become standard in trials of IBD drugs; such data are more relevant in clinical decision making and in crafting guidelines than placebo-based studies. In other recent head-to-head trials, guselkumab was superior to ustekinumab⁶, whereas mirikizumab was non-inferior to ustekinumab⁷. Finally, mounting evidence from these trials suggests that selective inhibition of IL-23, as with risankizumab and guselkumab (targeting the p19

Key advances

- The PROFILE study demonstrated that a top-down approach of infliximab plus immunomodulator was superior to accelerated step-up in the management of moderate-to-severe Crohn's disease².
- The head-to-head SEQUENCE study¹ showed that risankizumab was non-inferior to ustekinumab for inducing clinical remission in patients with moderate-to-severe Crohn's disease at week 24, and superior for endoscopic remission at week 48.
- In a healthy state, the *Bacteroides fragilis* polysaccharide A promoter is in the 'ON' orientation, enhancing the immunodulatory effects on regulatory T cells; this effect is lost when the gut mucosa is inflamed in experimental colitis, when there is a less diverse microbiota, and with greater exposure to bacteriophages³.

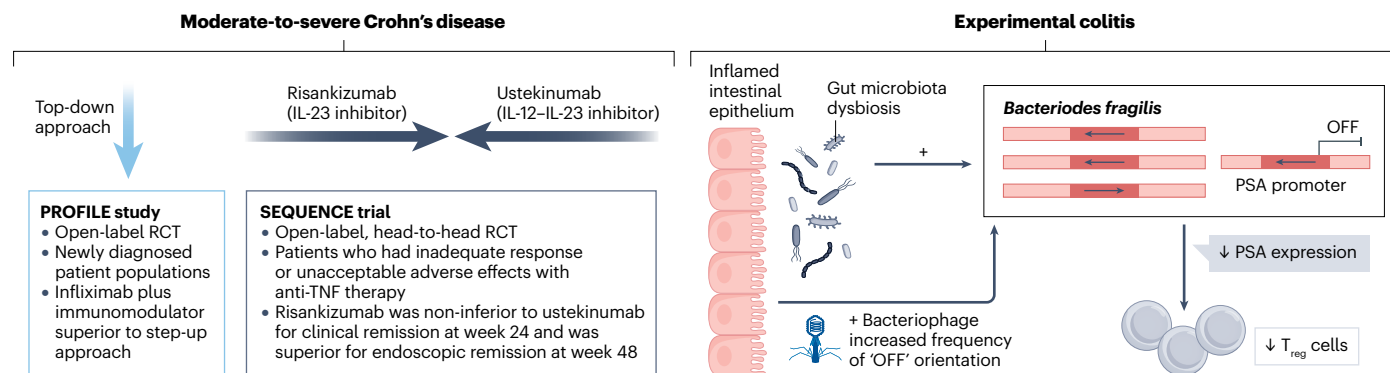


Fig. 1 | Advances in inflammatory bowel disease research in 2024. Clinical trials demonstrated the benefits of a top-down approach with early treatment (PROFILE study) and reported direct head-to-head comparisons of drugs targeting IL-12–IL-23 pathways (SEQUENCE trial). An experimental study

showed dynamic interactions between gut microbiota, bacteriophages and host inflammation. PSA, polysaccharide A; RCT, randomized controlled trial; T_{reg} cell, regulatory T cell.

subunit unique to IL-23), could be superior to dual inhibition of IL-12 and IL-23 by ustekinumab (targeting the shared p40 subunit). This new insight should guide further refinements of this drug class.

In 2024, the functional relationship between the gut microbiota, bacteriophages and host inflammatory response was explored³. In Bacteroidales, an abundant order of bacteria in the human gut, genes can be turned 'ON' or 'OFF' via inversion of DNA regions. For example, of eight capsular polysaccharides of the bacterium *Bacteroides fragilis*, seven have invertible promoters of the synthesis enzymes⁸. Switching polysaccharide synthesis ON or OFF can modulate the bacterial surface and can influence the interaction of polysaccharides with the host immune system. Polysaccharide A (PSA) of *B. fragilis* has known anti-inflammatory effects via the induction of regulatory T (T_{reg}) cells and other mechanisms⁹. A 2024 study unravelled the promotor inversion status of bacterial PSA in patients with IBD and mouse models with experimental colitis, versus healthy individuals and mice, and the role of bacteriophages³ (Fig. 1).

The invertible regions of Bacteroidales genomes from patients with IBD were analysed and showed that 147 regions had markedly different orientations compared with those in healthy individuals. This differential orientation was highly prevalent in proximity to the gene involved in capsular polysaccharide biosynthesis. In patients with IBD, a higher fraction of PSA promotors were in the OFF position than in healthy individuals. In follow-up experiments in mice transplanted with a healthy human microbiota, the *B. fragilis* PSA promotor increasingly changed from the ON to OFF orientation with onset of experimental colitis, and gradually returned to the ON orientation with resolution of inflammation. In additional experiments, *B. fragilis* exposed to faecal filtrates from patients with IBD before starting infliximab had an increased proportion of OFF PSA promotors; by contrast, the PSA promotor was mostly ON after exposure to faecal filtrates obtained from patients with IBD in whom inflammation had resolved after infliximab treatment. The researchers hypothesized that gut microbiota-derived bacteriophages might induce this differential PSA orientation in inflammation and found that 108 phages were associated with the *B. fragilis* PSA promotor OFF orientation. Finally, germ-free mice were monocolonized with *B. fragilis*, with and without the *B. fragilis* phage Barc2635 (isolated from sewage). In these mice, Barc2635 resulted in a higher fraction of the PSA promotor OFF direction, implying a lower expression of PSA on the bacterial surface. Strikingly, although *B. fragilis* monocolonization increased the fraction of FOXP3⁺ T_{reg} cells compared with germ-free mice, T_{reg} cells were reduced in the presence of phage Barc2635. Taken together, in intestinal inflammation, bacteriophages seem to turn the *B. fragilis* PSA promotor OFF, resulting in reduced PSA expression on the *B. fragilis* surface, depriving the host of the immunosuppressive effects of T_{reg} cells. The function

of the human intestinal virome is largely unknown, and unravelling the interplay between the highly diverse bacteriophage and bacterial populations and the host remains a major challenge, but has diagnostic and therapeutic potential.

2024 brought forth some practice-changing research. Early treatment for moderate-to-severe Crohn's disease was established as superior to a reactive approach, and these findings contributed to the latest treatment guidelines expunging the requirement for patients to have 'failed conventional treatment' before initiating biologic agents¹⁰. The hierarchy of efficacy of the anti-IL-12–IL-23 inhibitors became clearer, with selective IL-23 inhibition seemingly superior to dual IL-12–IL-23 blockade. However, medication choice will ultimately depend on the specific clinical situation, and the diverse, country-specific health funding policies. Finally, microbiome research moved forwards with more mechanistic insights, which could translate into new therapeutic approaches.

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Published online: 13 December 2024

References

- Peyrin-Biroullet, L. et al. Risankizumab versus ustekinumab for moderate-to-severe Crohn's disease. *N. Engl. J. Med.* **391**, 213–223 (2024).
- Noor, N. M. et al. A biomarker-stratified comparison of top-down versus accelerated step-up treatment strategies for patients with newly diagnosed Crohn's disease (PROFILE): a multicentre, open-label randomised controlled trial. *Lancet Gastroenterol. Hepatol.* **9**, 415–427 (2024).
- Carasso, S. et al. Inflammation and bacteriophages affect DNA inversion states and functionality of the gut microbiota. *Cell Host Microbe* **32**, 322–334.E9 (2024).
- Torres, J. et al. ECCO guidelines on therapeutics in Crohn's disease: medical treatment. *J. Crohns Colitis* **14**, 4–22 (2020).
- Narula, N. et al. Comparative effectiveness of biologics for endoscopic healing of the ileum and colon in Crohn's disease. *Am. J. Gastroenterol.* **117**, 1106–1117 (2022).
- Panaccione, R. et al. Efficacy and safety of guselkumab therapy in patients with moderately to severely active Crohn's disease: results of the galaxi 2 & 3 phase 3 studies. *Gastroenterology* **166**, 1057b1052 (2024).
- Jairath, V. et al. OP35 efficacy of mirikizumab in comparison to ustekinumab in patients with moderate to severe Crohn's disease: results from the phase 3 VIVID 1 study. *J. Crohns Colitis* **18**, i62–i64 (2024).
- Krinos, C. M. et al. Extensive surface diversity of a commensal microorganism by multiple DNA inversions. *Nature* **414**, 555–558 (2001).
- Mazmanian, S. K., Round, J. L. & Kasper, D. L. A microbial symbiosis factor prevents intestinal inflammatory disease. *Nature* **453**, 620–625 (2008).
- Gordon, H. et al. ECCO guidelines on therapeutics in Crohn's disease: medical treatment. *J. Crohns Colitis* **18**, 1531–1555 (2024).

Competing interests

The authors declare no competing interests.